

Statistical PDE learning

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Abstract

Abstract goes here.

1 Introduction: What is the problem?

The goal of the problem is as follows: the observation is (z_i, U_i) , where $z_i = (x_i, t_i) \in \Omega \times [0, \infty)$ is a spatial-time domain and $U_i = u(z_i) + \epsilon_i$ for $i = 1, \dots, n$ with $\epsilon_i \stackrel{i.i.d}{\sim} N(0, \sigma^2)$.

Suppose there exists a pre-defined dictionary $F_J := \text{span}\{f_1(u), \dots, f_J\}$. Our goal is to find a PDE equation whose solution is u . In particular, Brunton et al. (2016) considers the following task: assume

$$D^{\alpha^0} u = \sum_{s=1}^S D^{\alpha^s} g_s(u) = \sum_{s=1}^S \sum_{j=1}^J w_{(s-1)J+j} D^{\alpha^s} f_j(u), \quad g_s(u) \in F_J.$$

The following example will clarify this abstract notation.

Take f_j be the polynomial basis: $F_4 := \text{span}\{1, u, u^2, u^3\}$. And let $D^{\alpha^0} = I$ (zeroth order derivative), $D^{\alpha^1} = \partial_t$, $D^{\alpha^2} = \partial_x$, $D^{\alpha^3} = \partial_{xx}$. Suppose our target, true model has the following PDE:

$$u_t = 2\partial_t u^3 - \partial_x u^2 + 3\partial_{xx} u.$$

Then,

$$\begin{aligned} g_1(u) &= 2u^3 = 0 \cdot 1 + 0 \cdot u + 0 \cdot u^2 + 2 \cdot u^3, \\ g_2(u) &= -u^2 = 0 \cdot 1 + 0 \cdot u + -1 \cdot u^2 + 0 \cdot u^3, \\ g_3(u) &= 3u = 0 \cdot 1 + 3 \cdot u + 0 \cdot u^2 + 0 \cdot u^3. \end{aligned}$$

So, $w_1 = w_2 = w_3 = w_5 = w_6 = w_8 = w_9 = w_{11} = w_{12} = 0$, $w_4 = 2$, $w_7 = -1$, $w_{10} = 3$. This type of framework—modeling a target function by a span of a known basis—is called *dictionary learning* in numerical PDE literature. Typical example of dictionaries (F_J) includes polynomial basis, Fourier basis, and wavelet basis.

From now on, we will assume the true model exactly follows

$$D^{\alpha^0} u = \sum_{s,j} w_{(s-1)J+j}^* D^{\alpha^s} f_j(u). \tag{1}$$

In this case, finding the PDE equation boils down to finding $w_{(s-1)J+j}^*$.

2 What has been done before? Sindy, WSindy, and Wendy

In SINDY (Brunton et al., 2016), they first estimate $D^{\alpha^0} u$ and $D^{\alpha^s} f_j(u)$ from (z_i, U_i) . This needs to be done by replacing the derivative to the finite difference. This step is problematic, as estimating the differentials from the data already induces an error. This makes the regression problem ill-posed (both covariate and the response have errors, whose magnitude can be amplified as the differentiation order increases).

Keeping this significant bottleneck in our mind, SINDY learns $w_{s,j}^*$ by solving

$$\operatorname{argmin}_w \left\| \widehat{D^{\alpha^0} u}(z_i) - \sum_{s,j} w_{(s-1)J+j} \widehat{D^{\alpha^s} f_j}(u)(z_i) \right\|_2^2$$

possibly with the sparsity assumption on $w_{(s-1)J+j}^*$. In statistical language, $\widehat{D^{\alpha^s} f_j}(u)(z_i) = X_i \in \mathbb{R}^{S \times J}$, $\widehat{D^{\alpha^0} u}(z_i) = Y_i \in \mathbb{R}$, and we are estimating a coefficient w when the governing equation is $Y = X^T w^*$.

In the statistical viewpoint, there are several problems with this setup. To list a few, (i) the response variable and the covariate both contain errors—in particular, performing the differentiation amplifies the error, (ii) they are dependent as well, and (iii) the formulation is not doing the likelihood maximization (as it is not minimizing the noise).

WSINDY (Messenger and Bortz, 2021a) overcomes the issue of derivation in the response variable and the covariate by considering a weak form of the target equation.

To consider a weak form, for any test function (smooth, compactly supported function) ψ ,

$$D^{\alpha^0} u = \sum_{s,j} w_{(s-1)J+j}^* D^{\alpha^s} f_j(u) \Rightarrow \left\langle \psi, D^{\alpha^0} u \right\rangle_{L^2} = \left\langle \psi, \sum_{s,j} w_{(s-1)J+j}^* D^{\alpha^s} f_j(u) \right\rangle.$$

Using integration by parts and the fact that ψ has a compact support,

$$\left\langle (-1)^{|\alpha^0|} D^{\alpha^0} \psi, u \right\rangle = \sum_{s,j} w_{(s-1)J+j}^* \left\langle (-1)^{|\alpha^s|} D^{\alpha^s} \psi, f_j(u) \right\rangle \quad (2)$$

which is the form in Messenger and Bortz (2021a, Equation (3.2)). If (2) holds for every $\psi \in C_c^\infty(\Omega \times [0, \infty))$ and the integral is exact, then (1) also holds.

However, we neither can verify every test function nor compute the exact integral without the function form. Hence, two approximations are made. First, for the test function, Messenger and Bortz (2021a) chooses a fixed number of test functions $\{\psi_k\}_{k=1,\dots,K}$. Then, we want for $k = 1, \dots, K$,

$$\left\langle (-1)^{|\alpha^0|} D^{\alpha^0} \psi_k, u \right\rangle = \sum_{s,j} w_{(s-1)J+j}^* \left\langle (-1)^{|\alpha^s|} D^{\alpha^s} \psi_k, f_j(u) \right\rangle.$$

Next, for the L^2 -inner product,

$$\begin{aligned} \left\langle (-1)^{|\alpha^0|} D^{\alpha^0} \psi, u \right\rangle &\approx \sum_{i=1}^n (-1)^{|\alpha^0|} D^{\alpha^0} \psi_k(z_i) U_i \Delta z, \\ \left\langle (-1)^{|\alpha^0|} D^{\alpha^0} \psi, f_j(u) \right\rangle &\approx \sum_{i=1}^n (-1)^{|\alpha^s|} D^{\alpha^s} \psi_k(z_i) f_j(U_i) \Delta z. \end{aligned} \quad (3)$$

Hence, one solves

$$\underset{w}{\operatorname{argmin}} \sum_{k=1}^K \left\| \sum_{i=1}^n (-1)^{|\alpha^0|} D^{\alpha^0} \psi_k(z_i) U_i \Delta z - \sum_{s,j} w_{(s-1)J+j} \sum_{i=1}^n (-1)^{|\alpha^s|} D^{\alpha^s} \psi_k(z_i) f_j(U_i) \Delta z \right\|_2^2. \quad (4)$$

Again, in statistical languages, it is solving

$$R(w) := \hat{Y} - \hat{X}^T w \approx 0.$$

with

$$\hat{Y} = \operatorname{stack}_{k=1,\dots,K} \left(\sum_{i=1}^n (-1)^{|\alpha^0|} D^{\alpha^0} \psi_k(z_i) U_i \Delta z \right) \in \mathbb{R}^K$$

and

$$\hat{X} = \operatorname{stack}_{k,s,j} \left(\sum_{i=1}^n (-1)^{|\alpha^s|} D^{\alpha^s} \psi_k(z_i) f_j(U_i) \Delta z \right) \in \mathbb{R}^{K \times SJ}.$$

One clear advantage from SINDY is that we do not need to calculate (and cumulate the error) from differentiating U_i . All the differentials are taken w.r.t. test functions, which we can explicitly compute its derivatives.

Both SINDY and WSINDY adds a sparsity constraint to properly find the simple governing equation. Bertsimas and Gurnee (2023) uses a best subset selection idea from statistical literature (Bertsimas et al., 2016) to solve this.

One of the problem of SINDY and WSINDY is that the noise assumption is not the same as in the classical linear regression. To see this clear, in a typical regression setup, the source of the noise is directly a gap between fitted value and the target label, *i.e.*,

$$\epsilon_i = y_i - x_i^T \beta \stackrel{i.i.d}{\sim} N(0, \sigma^2),$$

while in SINDY and WSINDY, noise comes from $U_i = u(x_i) + \epsilon_i$, and $\hat{Y} - \hat{X}w$ is not merely a mean-zero noise, but more like a function of noises ϵ_i . WENDY (Bortz et al., 2023) tackles this problem. They decompose how the noise ϵ_i enters into the risk $R(w)$ and propose a method to adjust the effect of the noise.

For the notational simplicity, let

$$a_{s,k,i} := (-1)^{|\alpha^s|} D^{\alpha^s} \psi_k(z_i) \Delta z, \quad A_s := (a_{s,k,i})_{k,i} \in \mathbb{R}^{K \times n},$$

which is just to simply write

$$\hat{Y} = A_0 U, \quad \hat{X}_{\cdot, sj} = A_s f_j(U), \quad \hat{w} \in \underset{\|w\|_0 \leq q}{\operatorname{argmin}} \|R(w)\|_2^2 = \left\| A_0 U - \sum_{s,j} w_{(s-1)J+j} A_s f_j(U) \right\|_2^2. \quad (5)$$

Forget about the integration discretization error (this just reduces to the additive error). Then, $Y^* = A_0 u$, $X_{\cdot, sj}^* = A_s f_j(u)$. Our modeling assumption is

$$R^*(w^*) := A_0 u - \sum_{s,j} w_{(s-1)J+j}^* A_s f_j(u) = 0.$$

Now,

$$\hat{Y} - Y^* = A_0(U - u) = A_0\epsilon, \quad \hat{X} - X^* = \sum_{s,j} A_s (f_j(U) - f_j(u)).$$

Then,

$$\begin{aligned} R(w) &= \hat{Y} - \hat{X}w = (\hat{Y} - Y^*) - (\hat{X} - X^*)w + (Y^* - X^*w) \\ &= A_0\epsilon - \sum_{s,j} w_{s,j} A_s (f_j(U) - f_j(u)) + X^*(w^* - w). \end{aligned}$$

Now, $f_j(U) - f_j(u)$ part has hidden ϵ_i 's. Let's see whether we can explicitly write it. Since f_j can be in general non-linear, we will use Taylor expansion.

One important note: To perform a statistical theory analyses, it is natural to Taylor expand $f_j(U) = f_j(u) + f'_j(u)\epsilon + O(\epsilon^2)$, to decompose the population quantity and the noise. However, WENDY (Bortz et al., 2023) performs the opposite way— $f_j(u) = f_j(U) - f'_j(U)\epsilon + O(\epsilon^2)$. We will first summarize WENDY here, and then check what happens if we do the analyses in statistical perspective in Section 3.2.

To see what WENDY does, let's use

$$f_j(u) = f_j(U) + \text{diag}(f'_j(U_1), \dots, f'_j(U_n)) \epsilon + O(\epsilon^2).$$

Then,

$$R(w) = \left(A_0 - \sum_{s,j} w_{s,j} A_s \nabla f_j(U) \right) \epsilon + X^*(w^* - w) + \sum_{s,j} A_s O(\epsilon^2).$$

This formulation becomes clear once we make a comparison between the classical linear regression. In the classical linear regression,

$$\begin{aligned} R(\beta) &= Y - X\beta = X\beta^* + \epsilon - X\beta = X(\beta^* - \beta) + \epsilon \\ &\Rightarrow \epsilon = R(\beta^*). \end{aligned}$$

Hence,

$$R(w^*) = \left(A_0 - \sum_{s,j} w_{s,j}^* A_s \nabla f_j(U) \right) \epsilon + \sum_{s,j} A_s O(\epsilon^2)$$

is the regression *noise*. Note this new noise generally do not have a isotropic covariance even if ϵ has (ϵ_i 's may be correlated and heteroskedastic). Hence, Bortz et al. (2023) drops the $O(\epsilon^2)$ error, and then uses *generalized least squares* to fit $\hat{w}$. That said, WENDY (Bortz et al., 2023) does not provide a rigorous theoretical guarantee.

Messenger and Bortz (2025) shows that one can show the integration error can be ignored as data increases.

3 Issues in the previous methods

While WSINDY makes sense intuitively, there are several issues which hinders the statistical guarantee of WSINDY to hold.

Main issues can be summarized as follows:

1. The response variables and the covariates are approximations of the true integral, inducing an error. Messenger and Bortz (2025) shows this error can be removed when data densely covers the domain, with exponential convergence result, though the error analyses in the finite grid regime remains open.
2. Between $\hat{Y}$'s and $\hat{X}$'s, they are not independent (overlapping test functions gives the same information, and also they are weighted sum from the same raw data (z_i, U_i)), so that same ϵ_i appears simultaneously.
3. The error $\hat{Y} - \hat{X}^T w^*$ is not generally a mean-zero iid Gaussian. They can be correlated, heteroskedastic, non-Gaussian, and even can be non-zero if f_j is non-linear.
4. The finite collection of test functions determines the geometry of the population weak loss. Consequently, the residual directions being measured, identifiability, conditioning, and noise sensitivity depend on the choice of test functions. Its statistical interpretation, in particular, what the population target is, depends on the choice of the test function; this makes what the algorithm aims to solve elusive.

The above assessment gives one critical point we need to keep in mind for this problem, that is, there are two errors. First, there are errors induced from the noise (2 and 3). Second, there are errors that still appears even when there is no noise (1 and 4). Hence, it is good to separate these two, to understand the problem clearly.

Let's dig into each problem deeply.

3.1 Resolving the test function issue: Dual formulation

First of all, the choice of test function can be made more rigorously using the standard PDE regularity theory, as in Bonito et al. (2026).

The main message of Bonito et al. (2026) is that to learn the PDE solution correctly, one needs to choose the *correct* functional norm to measure the error for the given PDEs. For example, Bonito et al. (2026) claims that for the elliptic PDEs, H^1 norm is a natural choice to measure the error, as Lax-Milgram guarantees the solution to be in H^1 (Bonito et al., 2026, Equation (1.3)).

Unlike Bonito et al. (2026) where the goal is to learn the solution, our inverse problem learns the equation. Hence, there is no natural norm unlike Bonito et al. (2026).

However, one can *inversely* construct a natural functional norm $(\mathcal{X}, \|\cdot\|_{\mathcal{X}})$ once the dictionary F_J and differential operators $\{D^{\alpha^s}\}_{s=1,\dots,S}$ is fixed. Suppose the true model $u \in \mathcal{X}_0$, $f_j(u) \in \mathcal{X}_j$ for some Banach spaces $\mathcal{X}_j$. Such assumption is natural as dictionaries and the models are pre-defined. A standard duality inequality gives

$$\left\langle (-1)^{|\alpha^s|} D^{\alpha^s} \psi, f_j(u) \right\rangle \leq \|f_j(u)\|_{\mathcal{X}_j} \left\| (-1)^{|\alpha^s|} D^{\alpha^s} \psi \right\|_{\mathcal{X}_j^*}.$$

To make this inner-product a bounded operator, a naturally required condition is $\left\| (-1)^{|\alpha^s|} D^{\alpha^s} \psi \right\|_{\mathcal{X}_j^*} < \infty$.

Then, the natural choice of the norm is to bound these norms entirely. One canonical choice is

$$\|\psi\|_{\mathcal{X}}^2 := \|\psi\|_{L^p}^2 + \left\| D^{\alpha^0} \psi \right\|_{\mathcal{X}_0^*}^2 + \sum_{s,j} \left\| D^{\alpha^s} \psi \right\|_{\mathcal{X}_j^*}^2.$$

This is exactly the principal when constructing a Sobolev norm by the dual formulation. To get what this means in practice, a simple example is as follows: suppose

$$F_J := \text{span} \{u, u^2\}, \quad \{D^{\alpha^s}\}_{s=0,1,2} = \{I, \partial_x, \partial_{xx}\}.$$

Then, (1) consists of the following terms:

$$u, u^2, \partial_x u, \partial_x(u^2), \partial_{xx}(u), \partial_{xx}(u^2).$$

Now, suppose $u, u^2 \in L^2$. In the above abstract formulation, this means $\mathcal{X}_0 = \mathcal{X}_1 = \mathcal{X}_2 = L^2$. Then, for the inner-product in (2) to be well-defined, $\psi, \partial_x \psi, \partial_{xx} \psi \in (L^2)^* = L^2$. Hence, a natural norm is

$$\|\psi\|_{H^2}^2 = \|\psi\|_{L^2}^2 + \|\partial_x \psi\|_{L^2}^2 + \|\partial_{xx} \psi\|_{L^2}^2 < \infty.$$

Using this dual norm, we can avoid choosing K test functions. To this end, let's go back to (2). Let's define a residual functional

$$R(w)[\psi] := \left\langle (-1)^{|\alpha^0|} D^{\alpha^0} \psi, u \right\rangle - \sum_{s,j} w_{(s-1)J+j} \left\langle (-1)^{|\alpha^s|} D^{\alpha^s} \psi, f_j(u) \right\rangle.$$

$R(w^*)[\psi] = 0$ is exactly the constraint (2), and choosing K test functions means we reduce this constraint by $R(w^*)[\psi_k] = 0$ for $k = 1, \dots, K$.

A natural way to enforce (2) is to require

$$R(w^*)[\psi] = 0 \quad \forall \psi \in \mathcal{X} \quad \Leftrightarrow \quad \|R(w^*)\|_{\mathcal{X}^*} = 0.$$

In this perspective, a natural objective function to solve w now becomes

$$\hat{w} \in \underset{\|w\|_0 \leq q}{\operatorname{argmin}} \|R(w)\|_{\mathcal{X}^*} = \underset{\|w\|_0 \leq q}{\operatorname{argmin}} \sup_{\|\psi\|_{\mathcal{X}} \leq 1} \left| \left\langle (-1)^{|\alpha^0|} D^{\alpha^0} \psi, u \right\rangle - \sum_{s,j} w_{(s-1)J+j} \left\langle (-1)^{|\alpha^s|} D^{\alpha^s} \psi, f_j(u) \right\rangle \right|. \quad (6)$$

3.1.1 Choosing test function is equivalent to choosing the residual measuring norm

The above dual formulation gives the interpretation on the choice of the test function made in Messenger and Bortz (2021a). More precisely, the choice of the test function *leads to* the choice of the geometry we are measuring. In fact, while they did not make this interpretation, ODE version of WSINDY Messenger and Bortz (2021b) actually turned out to do the similar procedure.

Messenger and Bortz (2021b) can be re-interpreted as the dual norm minimization by the following procedure. Set

$$V_K := \text{span} \{\psi_1, \dots, \psi_K\}.$$

Now, define an inner product over V_K , provided it is well-defined:

$$\langle \psi_k, \psi_l \rangle_{V_K} := \sum_{i=1}^n \psi'_k(z_i) \psi'_l(z_i) (\Delta z)^2, \quad (G_K)_{k,l} := \langle \psi_k, \psi_l \rangle_{V_K}.$$

Now, they solve

$$\hat{w} \in \underset{w}{\operatorname{argmin}} R(w)^T G_K^{-1} R(w) + \text{regularizer} \quad (7)$$

which is just a weighted quadratic form instead of the plain squared loss.

On the other hand, inner product structure gives

$$R(w)^T G_K^{-1} R(w) = \sup_{\substack{\psi \in V_K \\ \|\psi\|_{V_K} \leq 1}} |R(w)[\psi]|^2.$$

And, if $K \rightarrow \infty$ and if ψ_i is chosen to be a basis of the Sobolev space H^1 , then

$$\langle \psi_k, \psi_l \rangle_{V_K} \xrightarrow{K \rightarrow \infty} \langle \psi_k, \psi_l \rangle_{H^1}.$$

Hence, (7) can be considered as a discretization of Sobolev dual norm minimization, *i.e.*,

$$\text{Population target of } \hat{w} \in \underset{w}{\operatorname{argmin}} \sup_{\substack{\psi \in H^1 \\ \|\psi\|_{H^1} \leq 1}} |R(w)[\psi]|^2 = \underset{w}{\operatorname{argmin}} \|R(w)\|_{H^{-1}}.$$

Not only this gives a justification, this shows the choice of the test function can be made systematically—we can choose which norm we want to minimize the residual at the beginning, and then the choice of the test function should be made to reflect it. In the above formulation, the choice of the inner product structure leads to measuring the residual error in H^{-1} norm.

3.2 Statistical analysis

We can choose whether we want to adopt the dual formulation (6), or stick with the finite choice of the test function (5). Integrating the dual formulation gives a more rigorous mathematics, but for now let's start with (5) for the simplicity.

There are multiple statistical analyses available based on our needs and attention. One can conduct non-asymptotic high-dimensional error analyses. One can do asymptotic analyses. One can model the noise distribution by Gaussian and do the distributional analyses.

3.2.1 Sanity check model: non-asymptotic error analysis of WSINDY

As an initial starting point, I will focus on how one can control $\|\hat{w} - w^*\|_2^2$ in a non-asymptotic sense. This is a standard target in the non-asymptotic high-dimensional regression. As mentioned, one can instead analyze asymptotics, consistency, distributional aspects, and so on. This is up to us.

What we will do in this section is to exactly mimic the high-dimensional linear regression techniques (Wainwright, 2019), and see where we get some flaws in the analyses. Recall the classical sparse linear regression, we set

$$\hat{\beta} \in \underset{\|\beta\|_0 \leq q}{\operatorname{argmin}} \|Y - X\beta\|_2^2.$$

A canonical condition in such analyses is *restricted curvature condition* (Wainwright, 2019). The condition assumes the existence of uniform $\kappa > 0$ such that

$$\frac{1}{n} \|Xv\|_2^2 \geq \kappa \|v\|_2^2 \quad \forall \|v\|_0 \leq 2q^1.$$

¹The displayed one is a simplified version, ignoring the tolerance term.

Restricted curvature condition yields

$$\left\| \hat{\beta} - \beta^* \right\|_2^2 \leq \frac{1}{\kappa n} \left\| X\hat{\beta} - X\beta^* \right\|_2^2 = \frac{1}{\kappa} R(\hat{\beta}). \quad (8)$$

We will see in a moment why this condition is favorable.

Now, we use the canonical analyses in high-dimensional regression and statistical learning theory, such as Wainwright (2019, Chapter 9). Starting from the basic inequality, we have

$$\begin{aligned} \frac{1}{n} \left\| Y - X\hat{\beta} \right\|_2^2 &\leq \frac{1}{n} \left\| Y - X\beta^* \right\|_2^2 \\ \Leftrightarrow \frac{1}{n} \left\| \epsilon - X(\hat{\beta} - \beta^*) \right\|_2^2 &\leq \frac{1}{n} \left\| \epsilon \right\|_2^2 \\ \Leftrightarrow \frac{1}{n} \left\| X(\hat{\beta} - \beta^*) \right\|_2^2 &\leq 2(\hat{\beta} - \beta^*)^T \frac{1}{n} X^T \epsilon. \end{aligned} \quad (9)$$

Apply (8) to the LHS. We get

$$\left\| \hat{\beta} - \beta^* \right\|_2^2 \leq \frac{2}{\kappa} (\hat{\beta} - \beta^*)^T \frac{X^T \epsilon}{n}. \quad (10)$$

Now, we only need to control the RHS. Standard Hölder inequality gives

$$(\hat{\beta} - \beta^*)^T \frac{X^T \epsilon}{n} \leq \left\| \hat{\beta} - \beta^* \right\|_1 \left\| \frac{X^T \epsilon}{n} \right\|_\infty.$$

Sparsity and Cauchy-Schwarz inequality yield

$$\left\| \hat{\beta} - \beta^* \right\|_1 \leq \sqrt{2q} \left\| \hat{\beta} - \beta^* \right\|_2.$$

Combining this to (10) yields

$$\left\| \hat{\beta} - \beta^* \right\|_2 \leq \frac{2\sqrt{2q}}{\kappa} \left\| \frac{X^T \epsilon}{n} \right\|_\infty. \quad (11)$$

This is a canonical form in sparse linear regression. The main analyses boil down to (i) finding κ , and (ii) bounding $\left\| X^T \epsilon / n \right\|_\infty$. In particular, controlling $\left\| X^T \epsilon / n \right\|_\infty$ using empirical process is what typically appears in the high-dimensional regression and statistical learning theory. This is typically done by

$$\left\| Z \right\|_\infty \leq \underbrace{\left\| Z - \mathbb{E}Z \right\|_\infty}_{\approx \text{variance}} + \underbrace{\left\| \mathbb{E}Z \right\|_\infty}_{\approx \text{bias}}.$$

We can essentially repeat the same analyses for WSINDY, and see how κ and $X^T \epsilon$ would look like. The restricted curvature condition, in a simpler form (ignoring general tolerance function), requires a uniform constant $\kappa > 0$ such that

$$\frac{1}{K} \left\| \hat{X}v \right\|_2^2 \geq \kappa \left\| v \right\|_2^2 \quad \forall \left\| v \right\|_0 \leq 2q.$$

In more precise form, this means

$$\frac{1}{K} \left\| \sum_{s,j} v_{(s-1)J+j} A_s f_j(U) \right\|_2^2 \geq \kappa \left\| v \right\|_2^2 \quad \forall \left\| v \right\|_0 \leq 2q.$$

The rest of analyses go the same. We in summary get

$$\|\hat{w} - w^*\|_2 \leq \frac{2\sqrt{2q}}{\kappa} \left\| \frac{\hat{X}^T R(w^*)}{K} \right\|_\infty. \quad (12)$$

As mentioned in the above, obtaining κ and controlling $\left\| \frac{\hat{X}^T R(w^*)}{K} \right\|_\infty$ are two bottlenecks in high-dimensional statistics. There are some problems with the current form, which hinders the analyses to go further:

1. κ : there is a stochasticity (more precisely, noise) involved in $f_j(U)$. Obtaining the analogous condition in stochastic sense will be needed. This should be possible using more general form of restricted curvature condition (including the tolerance term) (Wainwright, 2019), but need to check.
2. $\left\| \frac{\hat{X}^T R(w^*)}{K} \right\|_\infty$: There are several issues with this term. The precise term in the explicit notation is

$$\max_{s,j} \left| \frac{1}{K} (A_s f_j(U))^T \left(A_0 \epsilon - \sum_{s',j'} A_{s'} (f_{j'}(U) - f_{j'}(u)) \right) \right|.$$

- The bias may not be zero.
- Stochasticity is hard to analyze.

3.2.2 Integrating Section 3.1 and statistical analyses

In fact, if we assume the Hilbert space structure, as mentioned in the last part of Section 3.1, then the analyses of Section 3.2.1 carry over, as we can use the squared loss (or more generally a quadratic loss).

3.3 Computational aspects

TODO: Once we know what the *right* estimator to compute is, as we might need to modify the loss function itself.

4 Discussions

Misspecified model (dictionary) is deferred to later.

5 Space for general notes

General notes may go here...

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